

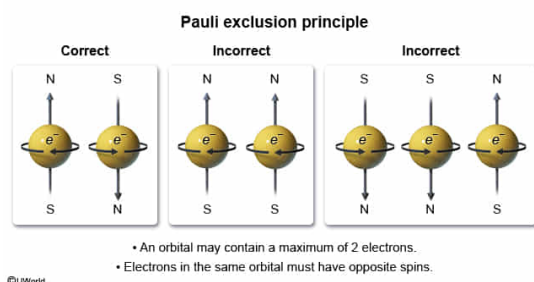
The spin quantum number m_s describes the quantum state of an electron in an orbital with respect to its angular momentum. According to the Pauli exclusion principle, what are the values of m_s for each of the electrons in the 4s orbital of a calcium atom?

- ☐ A. $+\frac{1}{2}$ and $+\frac{1}{2}$ [3%]
- ✓ ☒ B. $+\frac{1}{2}$ and $-\frac{1}{2}$ [94%]
- ☐ C. $-\frac{1}{2}$ and $-\frac{1}{2}$ [1%]
- ☐ D. $+\frac{1}{2}$, $-\frac{1}{2}$, and $+\frac{1}{2}$ [1%]

Correct

93% answered correctly

Explanation:



The quantum state of all electrons around an atom can be described using four quantum numbers (n , ℓ , m_ℓ , m_s):

- The principal quantum number n describes the main energy level (shell) of the electron and its most probable distance from the nucleus ($n = 1, 2, 3, 4, \dots$).
- The orbital angular momentum quantum number ℓ corresponds to the **subshell type** (s, p, d, f) and the corresponding **shape of the orbital** occupied by the electron, where $\ell = 0, 1, 2, 3, 4, \dots, (n - 1)$.
- The magnetic quantum number m_ℓ determines the number of orbitals of a given type and their **orientations** within a sublevel. For a given orbital, the values of m_ℓ include all integers ranging from $-\ell$ to $+\ell$.
- The electron spin quantum number m_s describes the angular momentum of an electron, which is a vector quantity with a magnitude of $\frac{1}{2}$ and a direction that is either **spin up (positive)** or **spin down (negative)**.

The Pauli exclusion principle states that each orbital subshell can hold a **maximum of two electrons**, but two electrons in the same orbital must have **opposite spins**. Therefore, the two electrons in the 4s orbital of a calcium atom must have opposite spins of $+\frac{1}{2}$ and $-\frac{1}{2}$.

(Choices A and C) Matching signs (+ or -) indicate matching electron spin orientations, and two electrons in the same orbital cannot have the same spin.

(Choice D) The 4s orbital holds only two (not three) electrons, so only two values of m_s would

be assigned (one for each electron).

Educational objective:

According to the Pauli exclusion principle, each orbital within a subshell can hold a maximum of two electrons, but two electrons in the same orbital must have opposite spins. Electron spin is a vector quantity with a magnitude of $\frac{1}{2}$ and a direction that is either spin up (positive) or spin down (negative).

Time spent:0 sec

Subject:
General Chemistry

QID:402740

System:
Atomic Theory and Chemical Composition

If a naturally occurring sample of an unidentified element is found to contain three isotopes (A, B, and C) and consists of 90.5% isotope A (mass number 20), 0.3% isotope B (mass number 21), and 9.2% isotope C (mass number 22), the atomic weight of the element measured from the sample will be:

- ☐ A. greater than 21 amu. [12%]
- ✓ ☒ B. less than 21 amu. [78%]
- ☐ C. equal to 21 amu. [7%]
- ☐ D. unrelated to the amount of isotope B. [1%]

Correct

78% answered correctly

Explanation:

Atomic weight of an element calculated from its natural isotope abundance

Isotope	Mass (amu)	Sample composition	Contribution to sample (amu)
A	20	90.5%	18.10
B	21	0.3%	0.06
C	22	9.2%	2.02
			20.18 amu

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Isotopes are atoms of the *same element* that have a different **number of neutrons** in the nucleus, and isotopes are identified by their **mass number** (the sum of the protons and neutrons). The masses of different isotopes are measured using the **atomic mass unit (amu)**, which is defined as one-twelfth of the mass contained in a carbon-12 atom (the average mass of 6 protons and 6 neutrons). However, in a **naturally occurring sample** of an element, a **mixture of isotopes** is present in percentages resulting from their respective **natural abundance**.

Because a naturally occurring sample of an element contains a mixture of isotopes, the atomic mass of a single isotope cannot be used in chemical calculations that involve bulk, macroscale samples. Instead, the **atomic weight** (more correctly called the relative atomic mass) is used, which accounts for the natural abundance of isotopes by taking a weighted contribution of the mass of each isotope found in the sample. It is these weighted averages that are listed as the atomic weights on the **periodic table**.

For the unidentified element in this question, the overwhelming majority of the sample is isotope A, which has a mass number of 20. Therefore, the atomic weight (relative atomic mass) measured for the sample will be closest to 20 amu because isotope A is

predominant within the sample.

(Choices A and C) Isotopes B and C together account for only 9.5% of the sample. Therefore, the relative atomic mass cannot be greater than or equal to 21 amu.

(Choice D) Although contributions from isotope B are small, they do have a slight impact on the value of the measured atomic weight.

Educational objective:

In a naturally occurring sample of an element, a mixture of isotopes is present in percentages resulting from their respective natural abundance. The atomic weights (relative atomic masses) listed on the periodic table account for natural isotope abundance by taking a weighted average of the mass of each isotope naturally found in a sample.

Time spent:0 sec

Subject:

General Chemistry

QID:402741

System:

Atomic Theory and Chemical Composition